Unified Modeling Language (UML) Structural & Behavioral Models

Academic Grievance, No-Dues & Clearance Automation Engine

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Submitted to: Dr. Rajesh Bhatia (Course: Software Engineering)

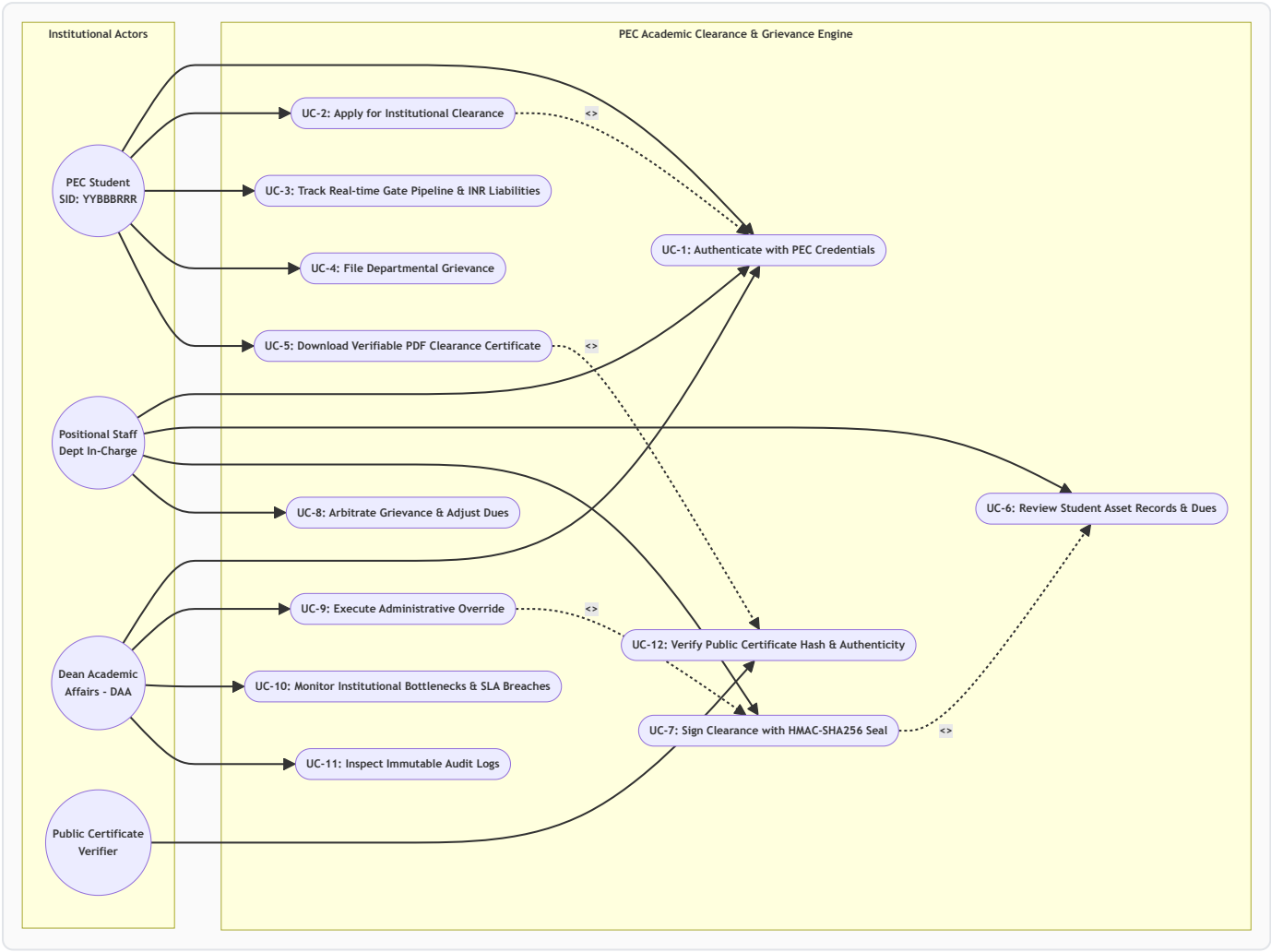
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1. Use Case Model & Diagram

1.1 Actors & Role Boundaries

- **Student Actor:** Registered PEC student identified by an 8-digit SID (e.g., `22103042`).
- **Positional Staff Actor:** Departmental authority operating under permanent institutional designations (*Librarian*, *Chief Warden*, *Gymkhana President*, *HOD Labs*, *Accounts Officer*).
- **Dean Academic Affairs (DAA / Admin Actor):** Executive authority holding override privileges and institutional audit oversight.
- **Public Certificate Verifier Actor:** External unauthenticated party validating clearance credentials.

1.2 Use Case Diagram



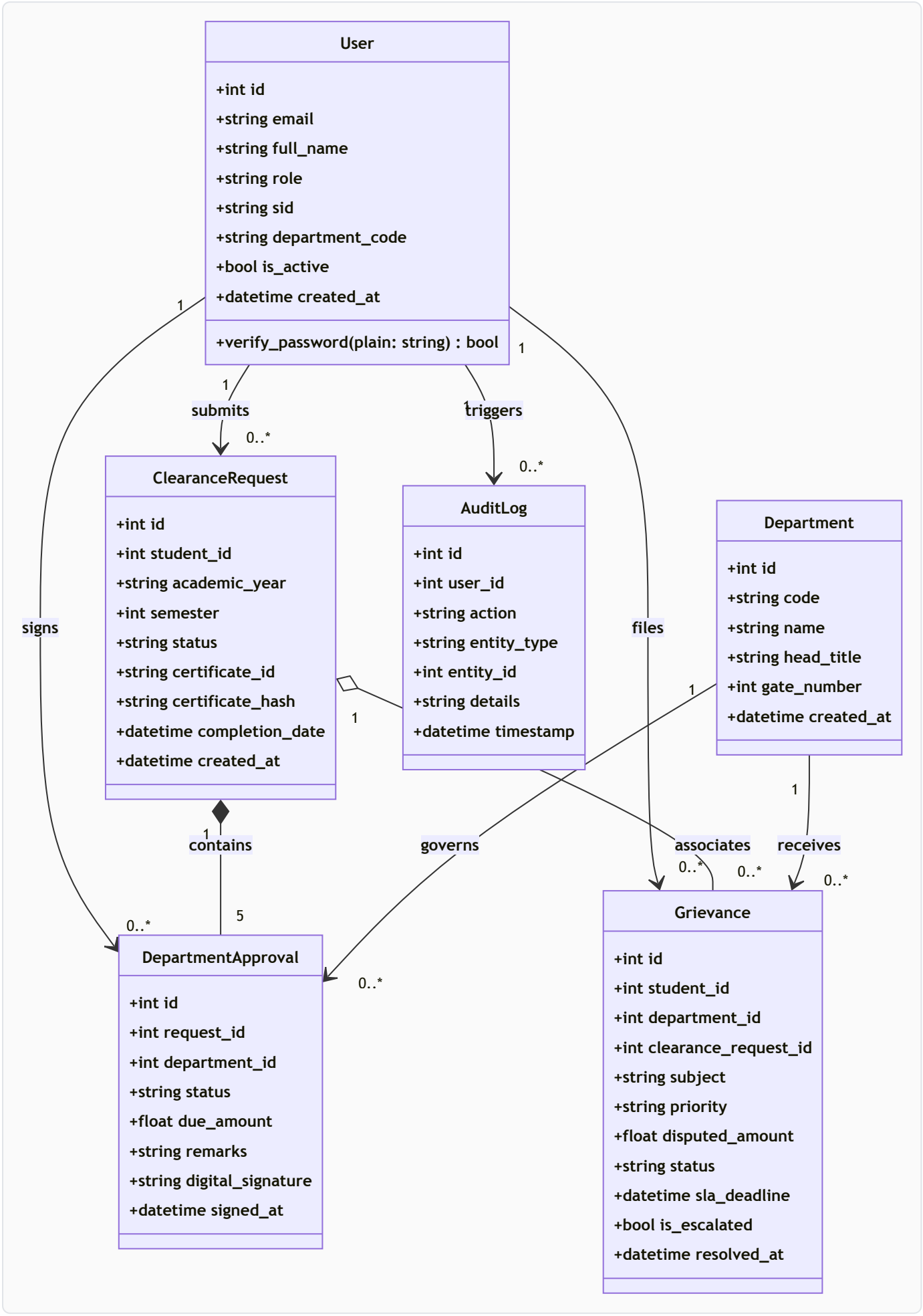
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2. Static Structural Model: Class Diagram

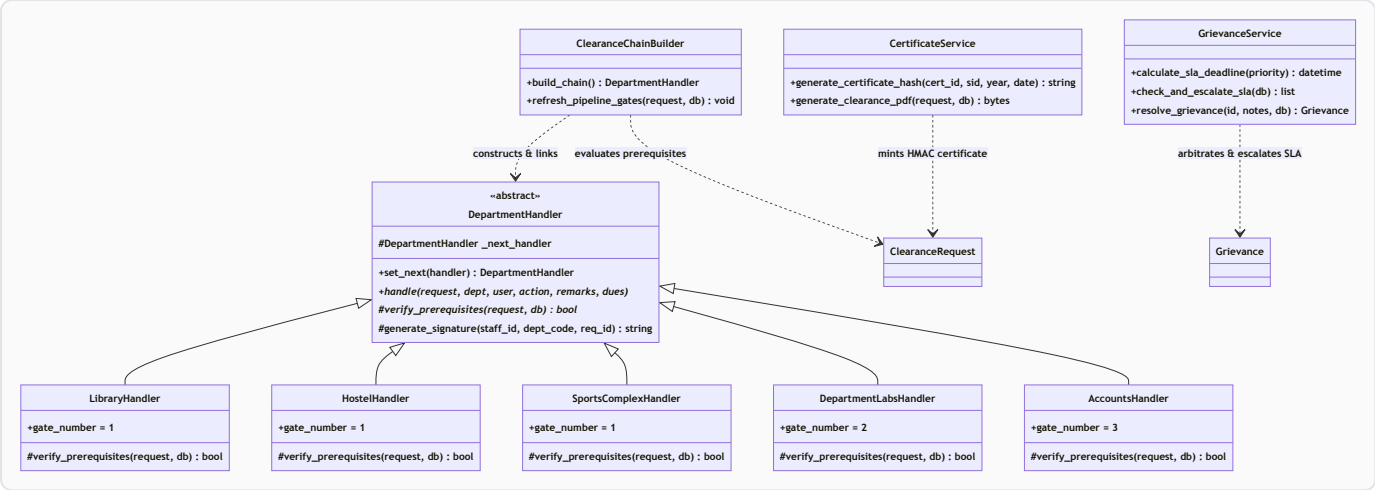
The static structural architecture of the system is divided into two focused, high-contrast class diagrams to ensure maximum visual clarity and readability:

1. **Core Domain Entity Model** (relational data models, attributes, methods, associations, and multiplicities)
2. **Behavioral Chain of Responsibility & Services Model** (handler hierarchy, gate verification, and service orchestration)

2.1 Core Domain Entity Class Diagram


User
+int id
+string email
+string full_name
+string role
+string sid
+string department_code
+bool is_active
+datetime created_at
+verify_password(plain: string) : bool
ClearanceRequest
+int id
+int student_id
+string academic_year
+int semester
+string status
+string certificate_id
+string certificate_hash
+datetime completion_date
+datetime created_at
AuditLog
+int id
+int user_id
+string action
+string entity_type
+int entity_id
+string details
+datetime timestamp
Department
+int id
+string code
+string name
+string head_title
+int gate_number
+datetime created_at
DepartmentApproval
+int id
+int request_id
+int department_id
+string status
+float due_amount
+string remarks
+string digital_signature
+datetime signed_at
Grievance
+int id
+int student_id
+int department_id
+int clearance_request_id
+string subject
+string priority
+float disputed_amount
+string status
+datetime sla_deadline
+bool is_escalated
+datetime resolved_at
submits
triggers
signs
files
contains
governs
associates
receives
1
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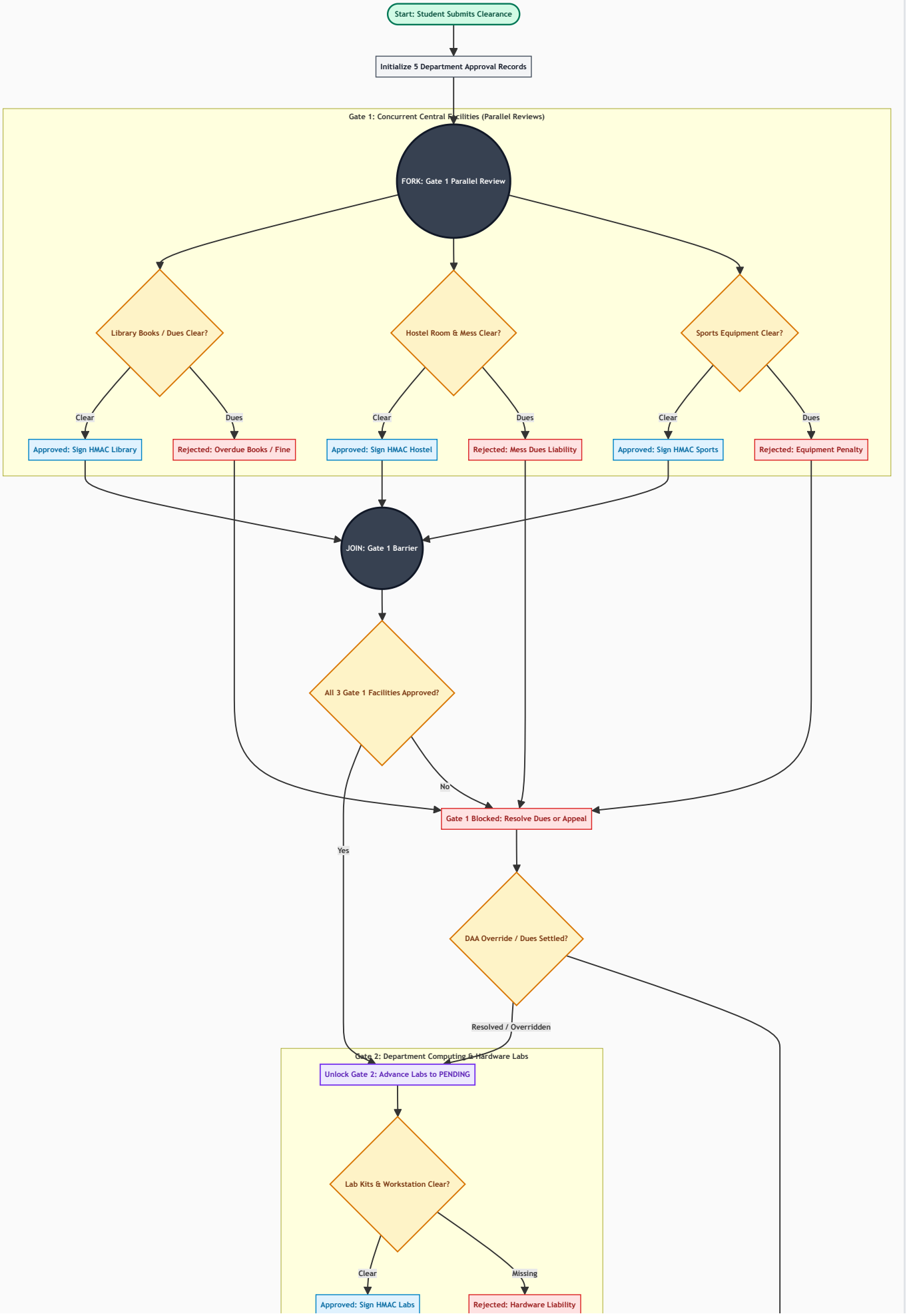
2.2 Behavioral Chain of Responsibility & Service Orchestration Diagram

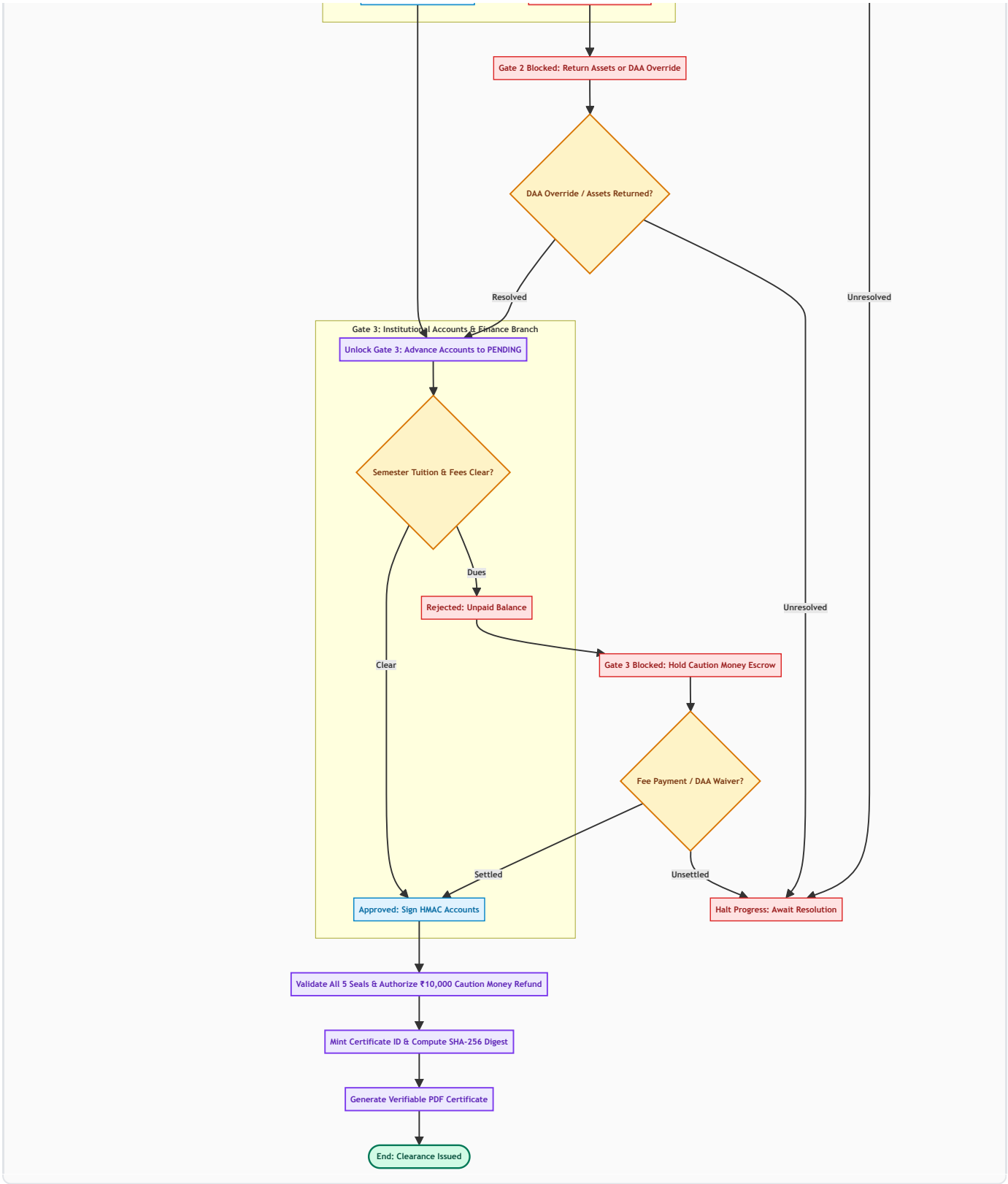


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3. Dynamic Behavioral Model: Activity Diagram

This activity diagram models the end-to-end dynamic workflow of a PEC clearance request, illustrating parallel concurrency forks, barrier joins, and sequential dependency gates:


Start: Student Submits Clearance
Initialize 5 Department Approval Records
Gate 1: Concurrent Central Facilities (Parallel Reviews)
FORK: Gate 1 Parallel Review
Library Books / Dues Clear?
Hostel Room & Mess Clear?
Sports Equipment Clear?
Clear
Dues
Clear
Dues
Clear
Dues
Approved: Sign HMAC Library
Rejected: Overdue Books / Fine
Approved: Sign HMAC Hostel
Rejected: Mess Dues Liability
Approved: Sign HMAC Sports
Rejected: Equipment Penalty
JOIN: Gate 1 Barrier
All 3 Gate 1 Facilities Approved?
No
Yes
Gate 1 Blocked: Resolve Dues or Appeal
DAA Override / Dues Settled?
Resolved / Overridden
Gate 2: Department Computing & Hardware Labs
Unlock Gate 2: Advance Labs to PENDING
Lab Kits & Workstation Clear?
Clear
Missing
Approved: Sign HMAC Labs
Rejected: Hardware Liability

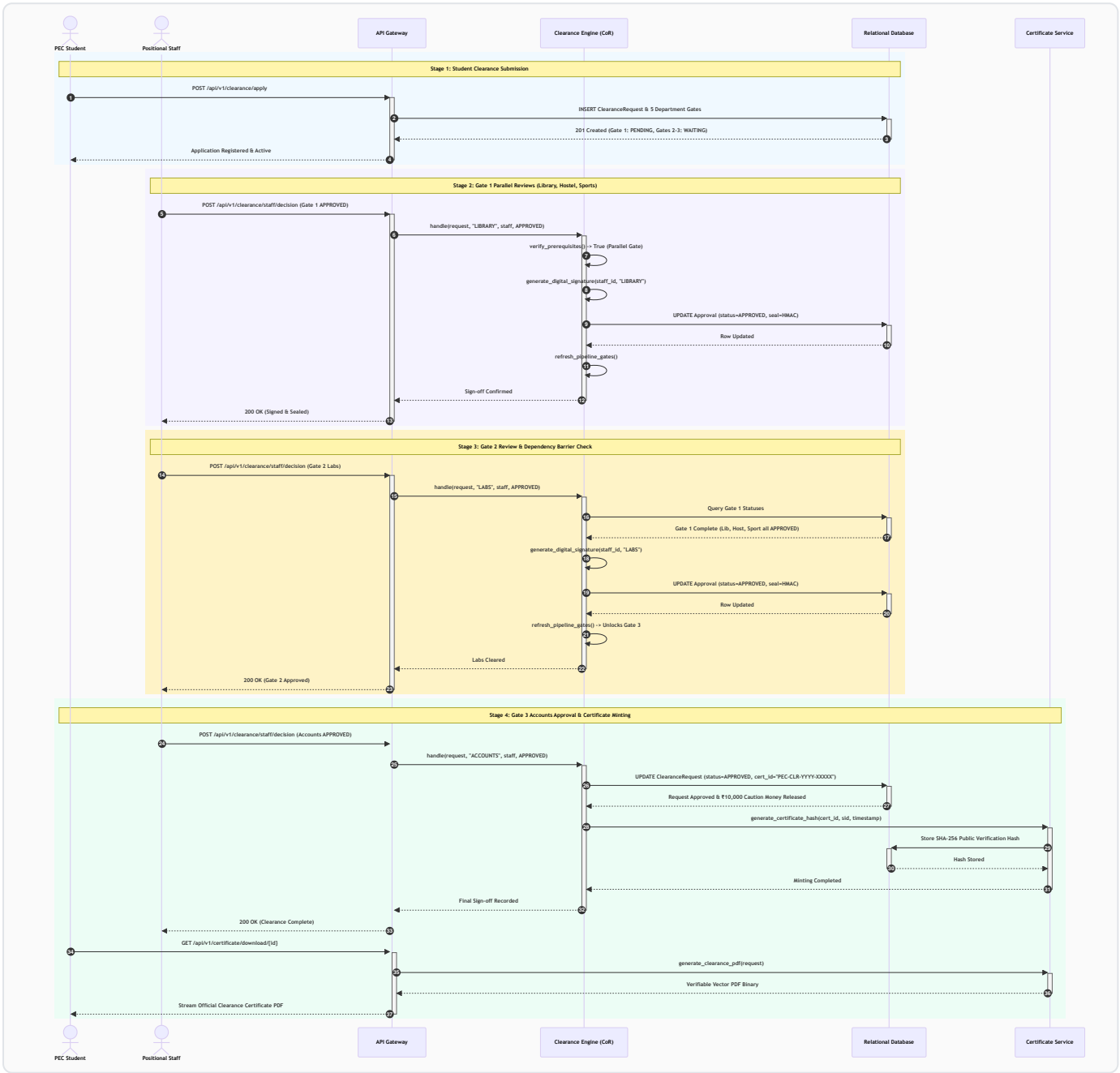


4. Interaction Models: Sequence Diagrams

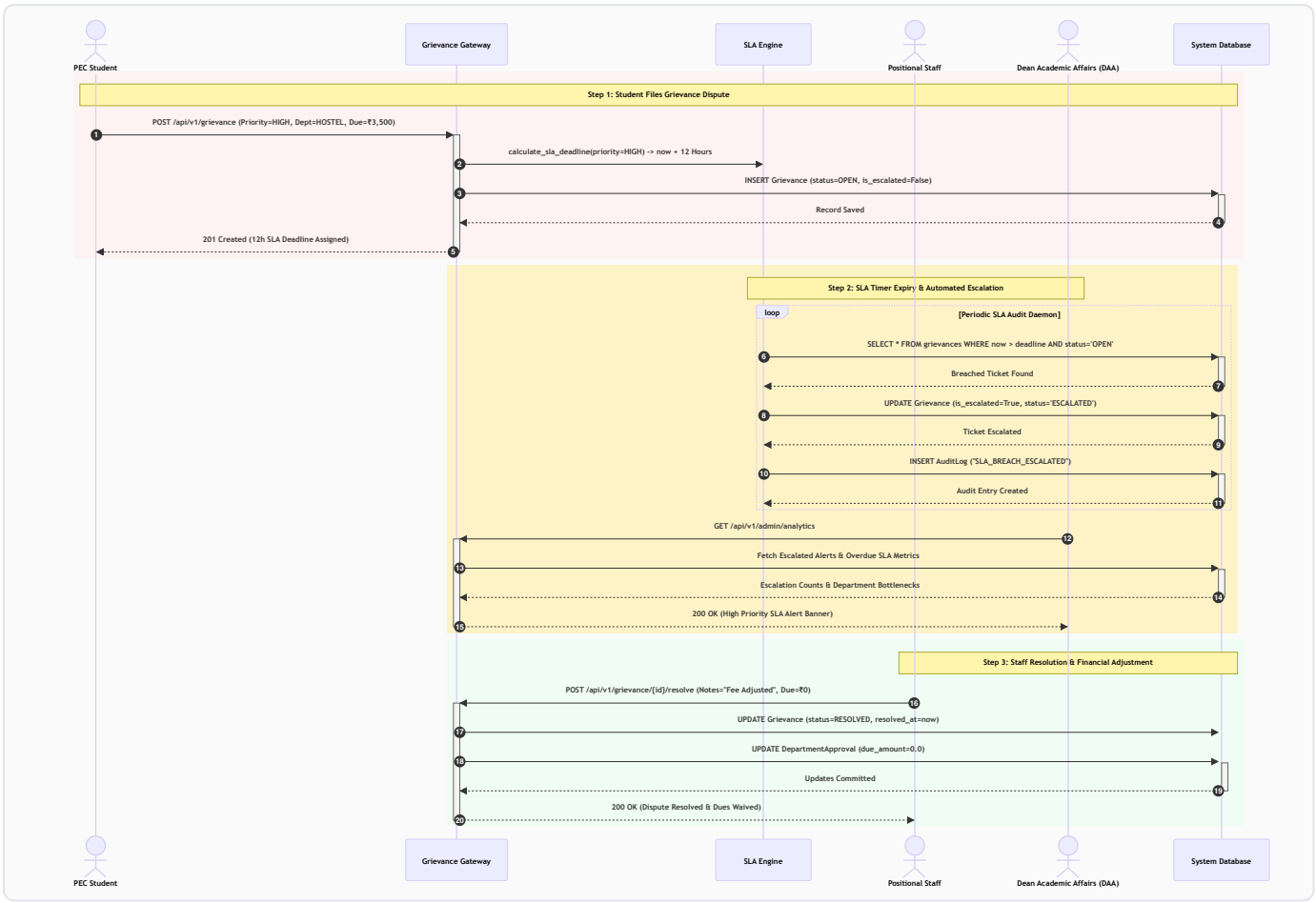
The dynamic interaction between external actors, API gateways, business logic engines, and database layers is presented across three structured, chronological sequence diagrams:

1. **Multi-Gate Approval & Cryptographic Minting** (Clearance lifecycle, prerequisite gating, HMAC sealing, and PDF issuance)
2. **Grievance Dispute Lifecycle & Automated SLA Escalation** (Student appeals, timer countdowns, and DAA alerts)
3. **Executive Administrative Override & Serialization** (DAA barrier unlocks, audit trail logging, and concurrency control)

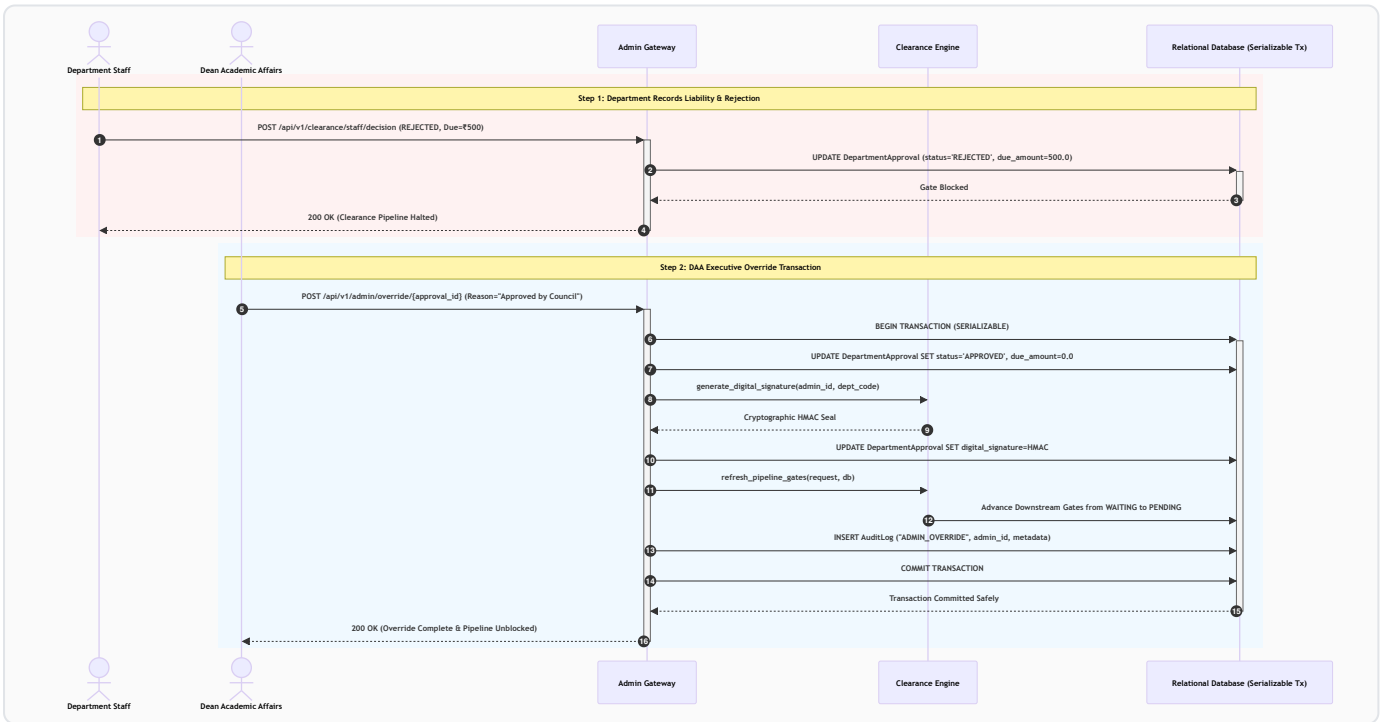
4.1 Sequence Diagram 1: Multi-Gate Approval & Cryptographic Minting



4.2 Sequence Diagram 2: Grievance Dispute Lifecycle & Automated SLA Escalation

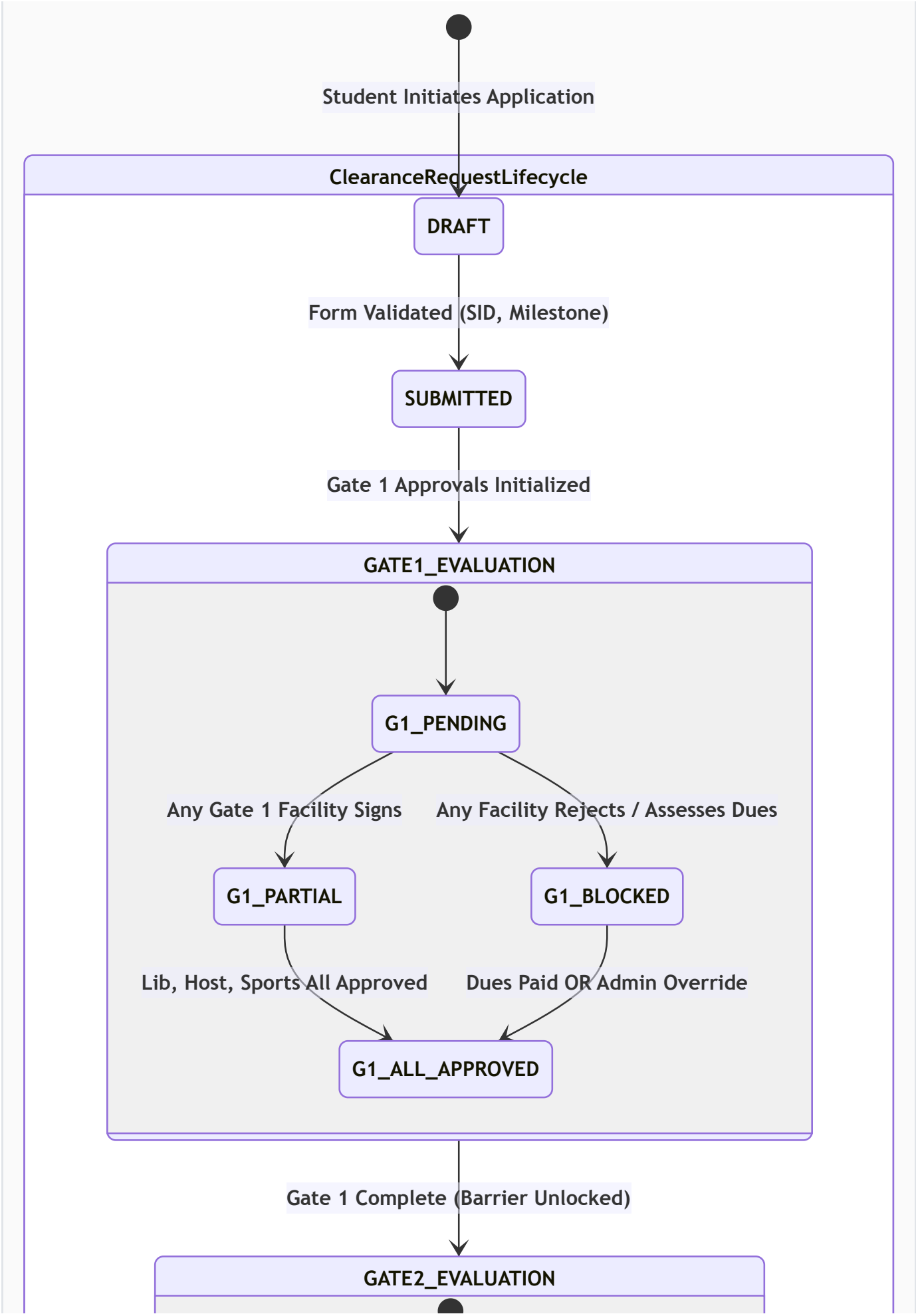


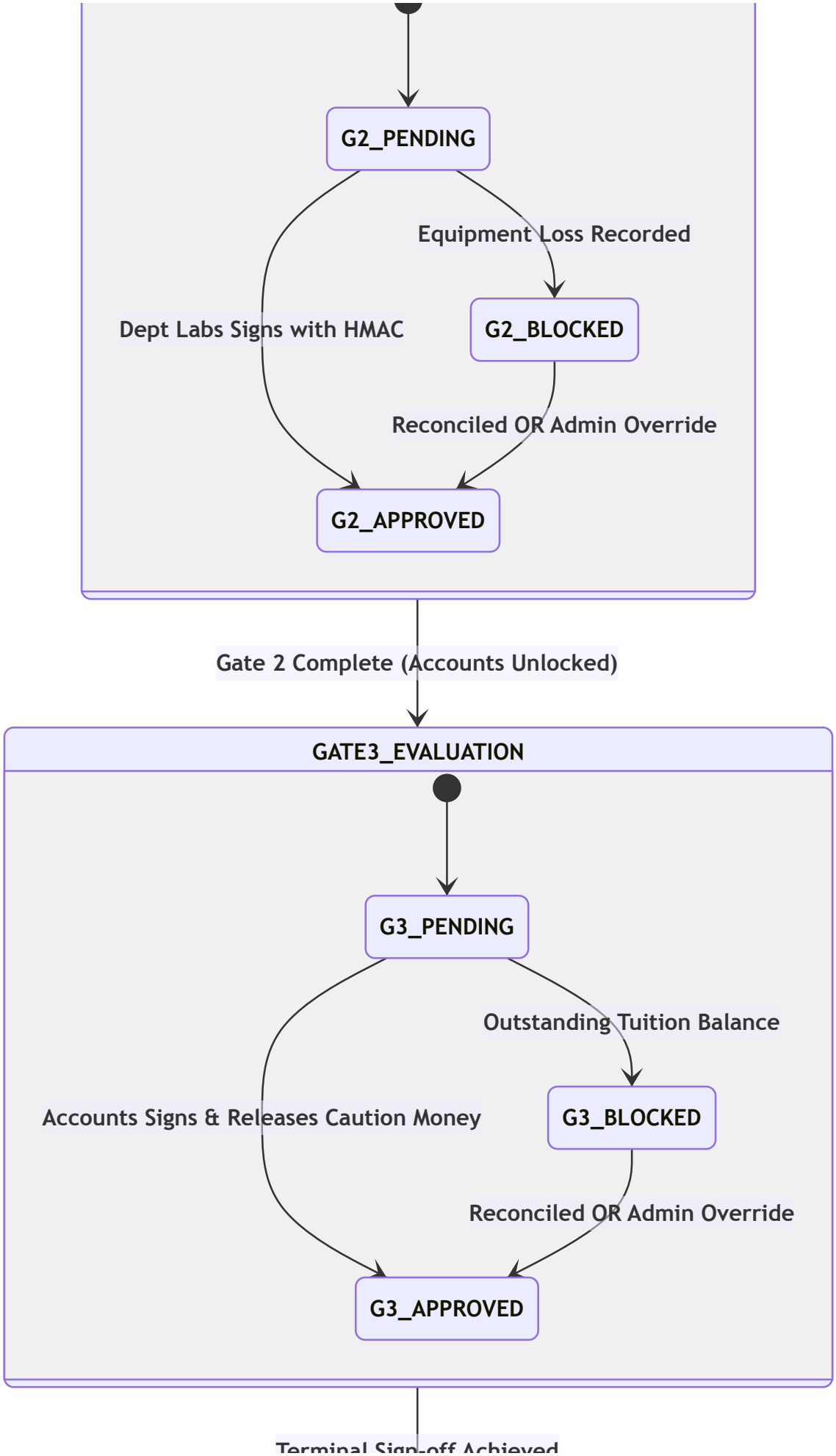
4.3 Sequence Diagram 3: Administrative Override & Concurrency Control

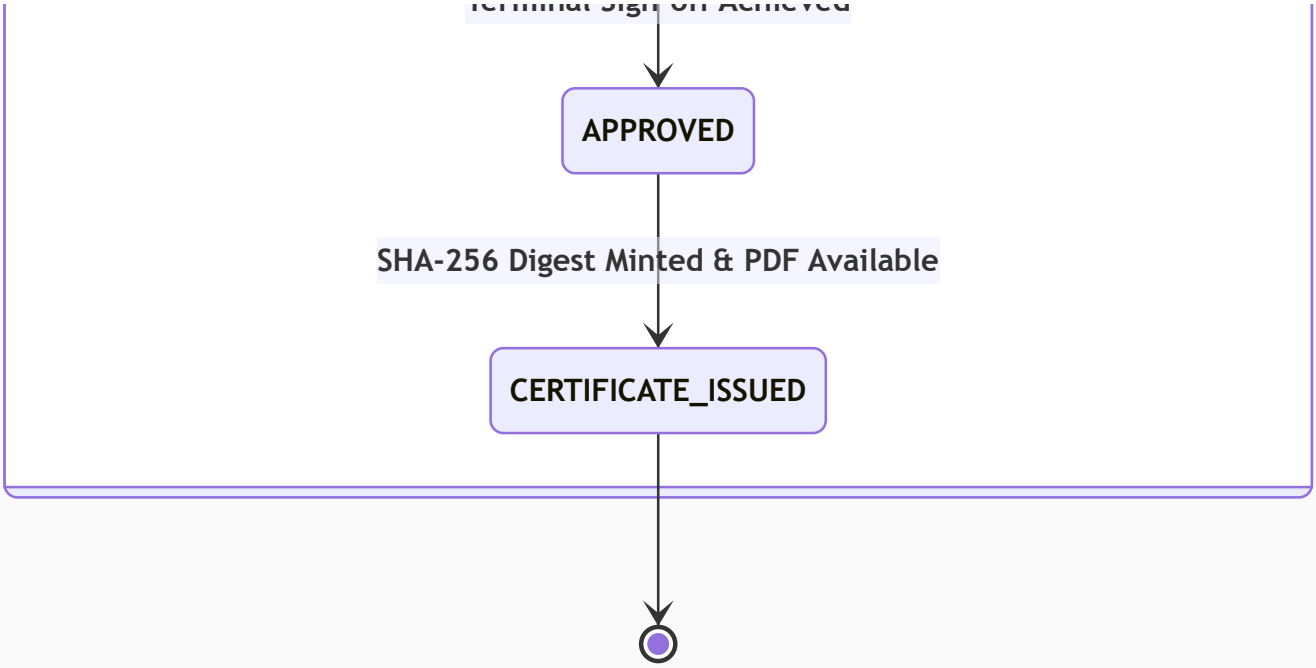


5. Behavioral State Machine Diagram

This state machine models the discrete status lifecycles of both the master clearance request and individual department approval gates:

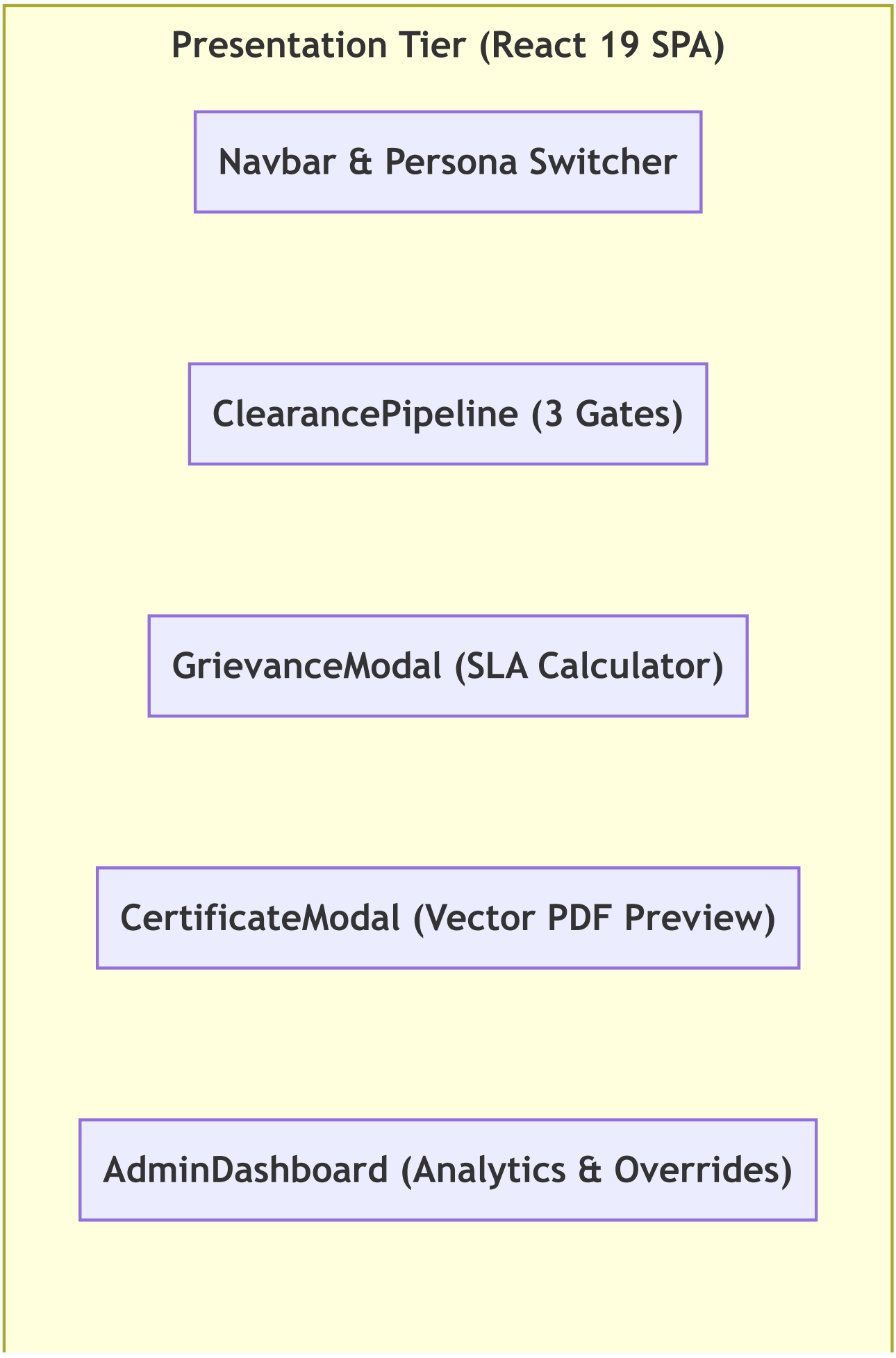

Student Initiates Application
ClearanceRequestLifecycle
DRAFT
Form Validated (SID, Milestone)
SUBMITTED
Gate 1 Approvals Initialized
GATE1_EVALUATION
G1_PENDING
Any Gate 1 Facility Signs
Any Facility Rejects / Assesses Dues
G1_PARTIAL
G1_BLOCKED
Lib, Host, Sports All Approved
Dues Paid OR Admin Override
G1_ALL_APPROVED
Gate 1 Complete (Barrier Unlocked)
GATE2_EVALUATION

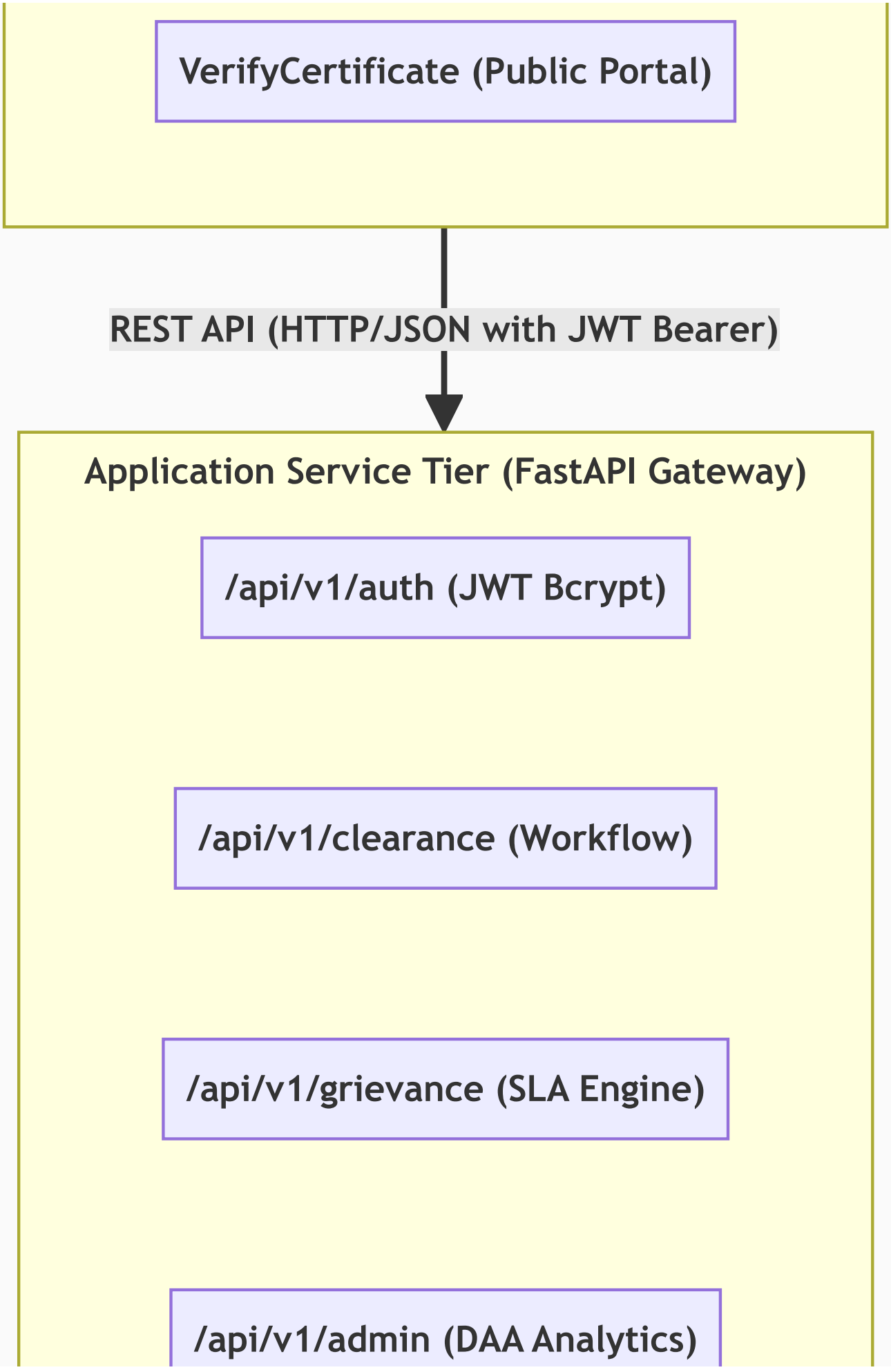

G2_PENDING
Equipment Loss Recorded
Dept Labs Signs with HMAC
G2_BLOCKED
Reconciled OR Admin Override
G2_APPROVED
Gate 2 Complete (Accounts Unlocked)
GATE3_EVALUATION
G3_PENDING
Outstanding Tuition Balance
Accounts Signs & Releases Caution Money
G3_BLOCKED
Reconciled OR Admin Override
G3_APPROVED
Terminal Sign-off Achieved

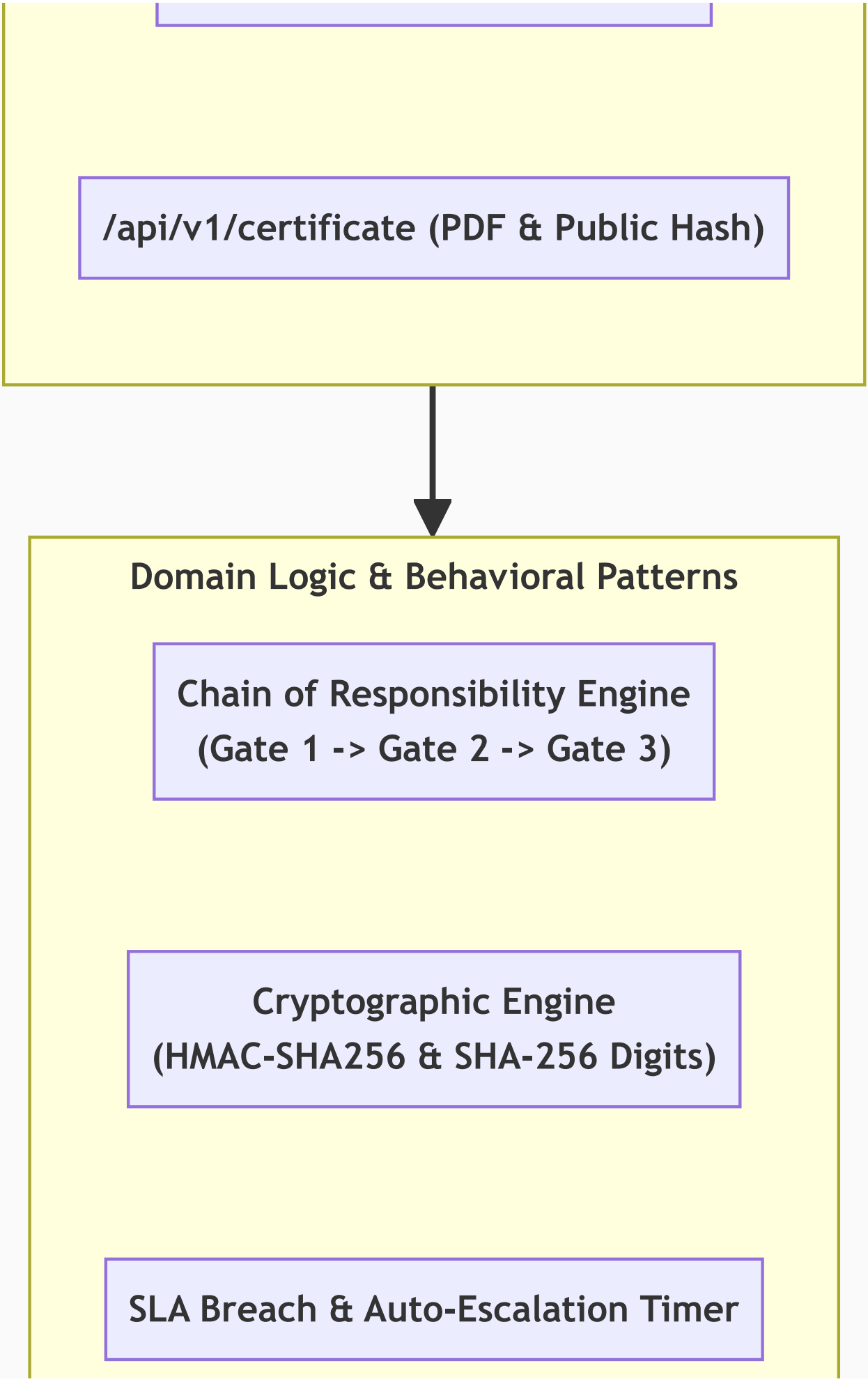


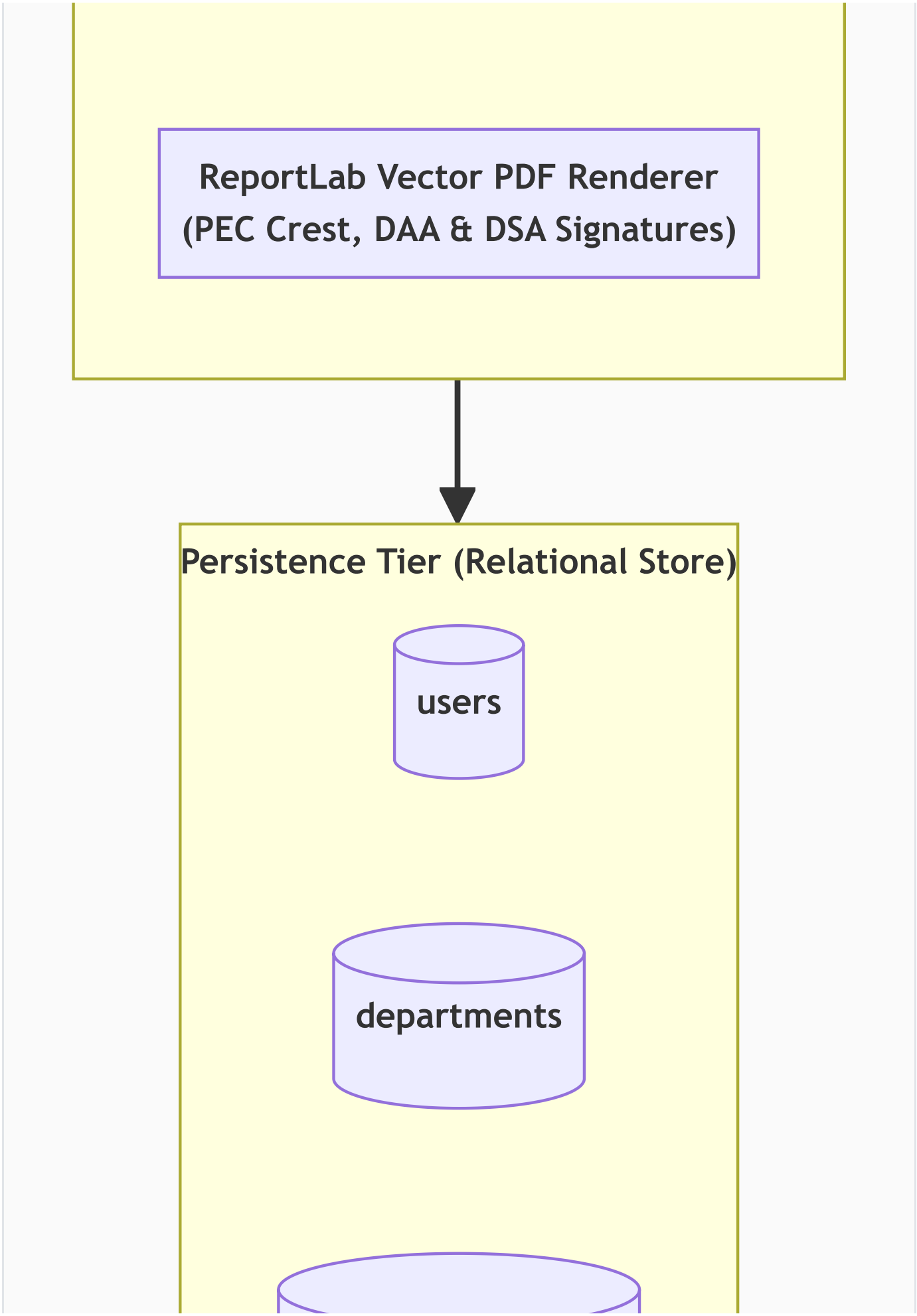
6. System Architecture & Component Deployment Diagram

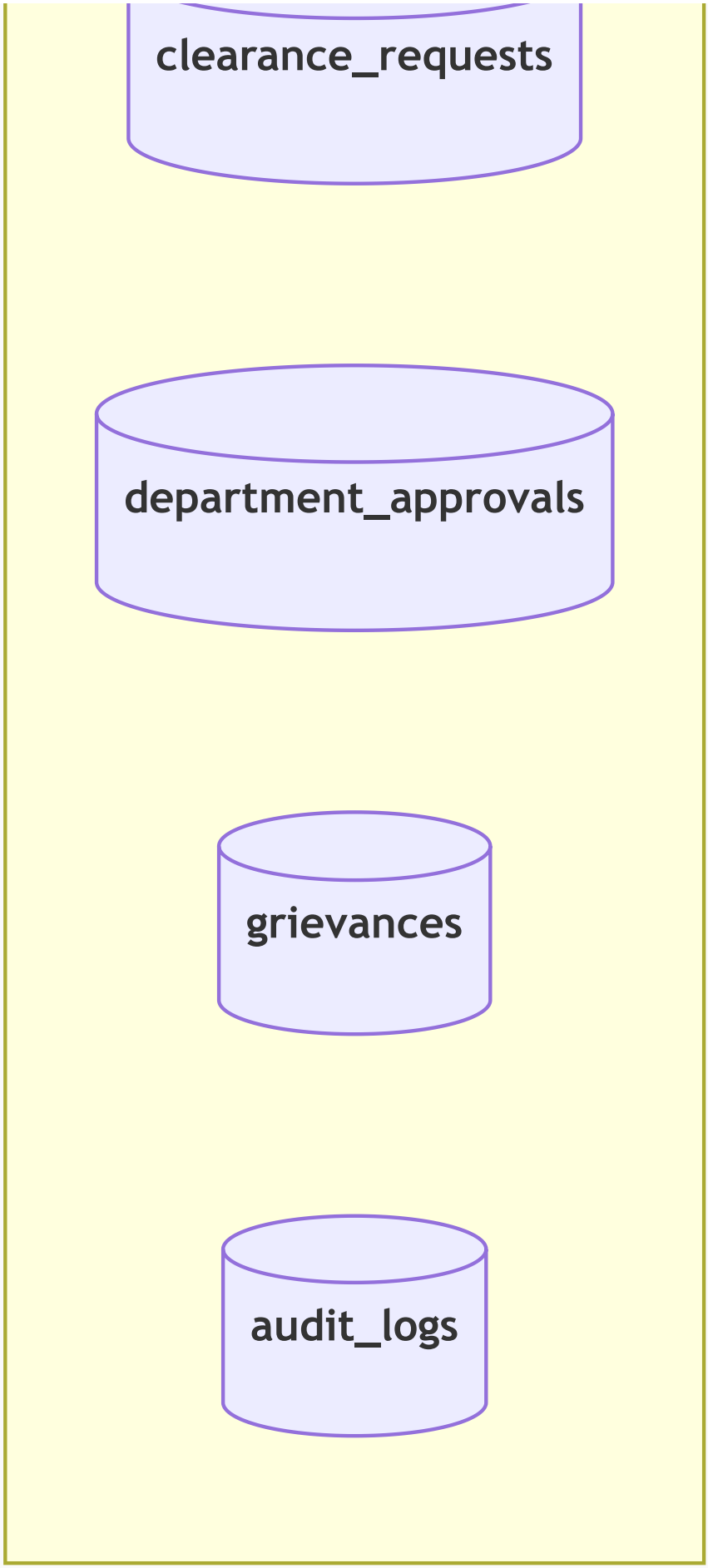
The component diagram defines the physical and logical boundaries of the system, verifying complete separation between presentation, application services, and persistent storage:


Presentation Tier (React 19 SPA)
Navbar & Persona Switcher
ClearancePipeline (3 Gates)
GrievanceModal (SLA Calculator)
CertificateModal (Vector PDF Preview)
AdminDashboard (Analytics & Overrides)


VerifyCertificate (Public Portal)
REST API (HTTP/JSON with JWT Bearer)
Application Service Tier (FastAPI Gateway)
/api/v1/auth (JWT Bcrypt)
/api/v1/clearance (Workflow)
/api/v1/grievance (SLA Engine)
/api/v1/admin (DAA Analytics)


/api/v1/certificate (PDF & Public Hash)
Domain Logic & Behavioral Patterns
Chain of Responsibility Engine
(Gate 1 -> Gate 2 -> Gate 3)
Cryptographic Engine
(HMAC-SHA256 & SHA-256 Digits)
SLA Breach & Auto-Escalation Timer


ReportLab Vector PDF Renderer
(PEC Crest, DAA & DSA Signatures)
Persistence Tier (Relational Store)
users
departments


clearance_requests
department_approvals
grievances
audit_logs